Lecture-25
Internal Flows IV: Energy considerations in pipe flow

Saikat Mukherjee

Department of Mechanical Engineering, IIT Kharagpur

Learning Objectives

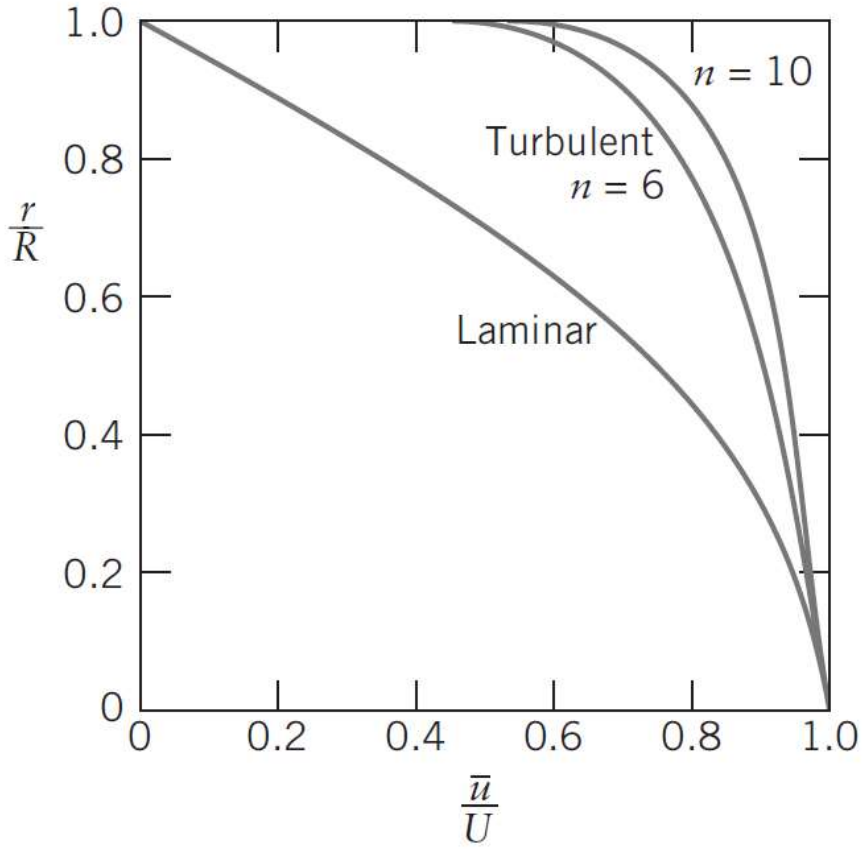
By the end of this lecture, you should be able to

1. Derive the **pipe flow energy equation** by applying the Reynolds Transport Theorem to internal energy
2. Quantify **irreversible energy losses** from the derived energy balance
3. Define the **friction factor** used to calculate head loss in piping systems

Power law equation for flow in a pipe

- Turbulent velocity profile inside a pipe may be approximated using an **empirical power-law** equation

$$\frac{\bar{u}}{U} = \left(\frac{y}{R}\right)^{1/n} = \left(1 - \frac{r}{R}\right)^{1/n}$$



- This is **not applicable for flow near the wall**, it gives infinite shear stress !!
- At the centerline, it **does not give a perfectly zero slope,** but the profile shows a decent fit otherwise.
- The value of the power law coefficient varies with Reynolds number

$$n = -1.7 + 1.8 \log Re_U$$

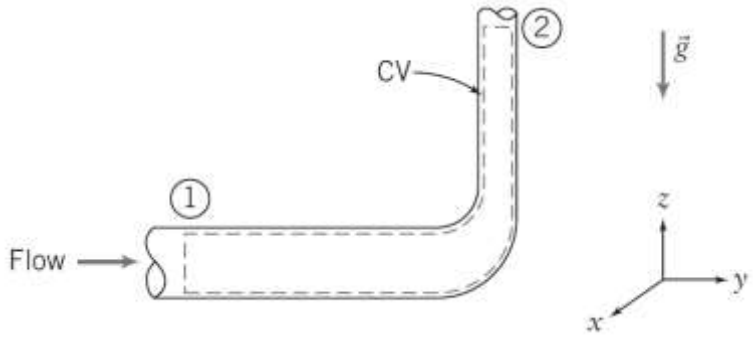
Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Energy considerations in pipe flow

- Reynolds transport theorem for **conservation of energy** gives us a baseline for energy considerations in a pipe flow.

$$\dot{Q} = \dot{m}(u_2 - u_1) + \dot{m}\left(\frac{p_2}{\rho} - \frac{p_1}{\rho}\right) + \dot{m}g(z_2 - z_1)$$

$$+ \int_{A_2} \frac{V_2^2}{2} \rho V_2 \, dA_2 - \int_{A_1} \frac{V_1^2}{2} \rho V_1 \, dA_1$$



- Kinetic energy coefficient can be used to simplify the integrals

$$\int_A \frac{V^2}{2} \rho V \, dA = \alpha \int_A \frac{\overline{V}^2}{2} \rho V dA = \alpha \dot{m} \frac{\overline{V}^2}{2}$$

- The equation can be rearranged to give

$$\left(\frac{p_1}{\rho} + \alpha_1 \frac{\overline{V}_1^2}{2} + g z_1\right) - \left(\frac{p_2}{\rho} + \alpha_2 \frac{\overline{V}_2^2}{2} + g z_2\right) = (u_2 - u_1) - \frac{\delta Q}{dm}$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Calculation of head loss

- When the **elevation is constant**, and flow is **fully developed**, head losses can be attributed to the **loss of pressure head**

$$\frac{p_1 - p_2}{\rho} = \frac{\Delta p}{\rho} = h_l$$

- When the flow is **laminar**, we have an analytical expression for pressure drop

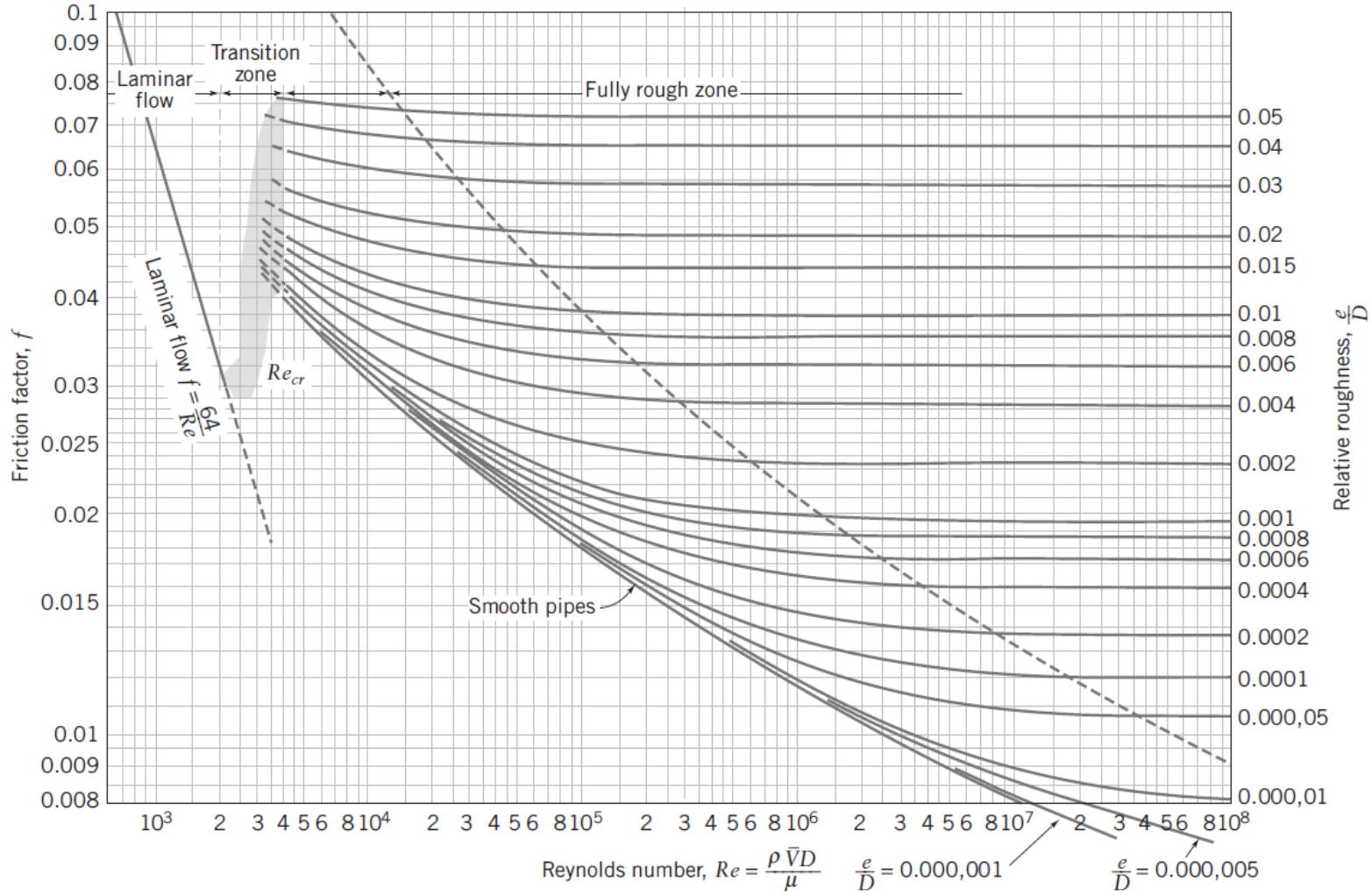
$$h_l = 32 \frac{L}{D} \frac{\mu \overline{V}}{\rho D} = \frac{L}{D} \frac{\overline{V}^2}{2} \left(64 \frac{\mu}{\rho \overline{V} D} \right) = \left(\frac{64}{Re} \right) \frac{L}{D} \frac{\overline{V}^2}{2}$$

- When the flow is **turbulent**, we use **dimensional analysis** to simplify the problem

$$h_l = f \frac{L}{D} \frac{\overline{V}^2}{2}$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Calculation of head loss – Moody charts



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

1. Derive the **pipe flow energy equation** by applying the Reynolds Transport Theorem to internal energy
2. Quantify **irreversible energy losses** from the derived energy balance
3. Define the **friction factor** used to calculate head loss in piping systems

